

Adaptive Importance Sampling

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Outline

- Importance sampling
- Measuring performance
 - Effective sample size
- Improving performance
 - Stochastic approximation

Importance Sampling

- Need to compute an expected value
 - $\mathbb{E}_F \varphi(X) =: \mathcal{I}$
- Can't do the sum/integral
- Monte Carlo approximation
 - Simulating from F might be hard

Importance Sampling

- Introduce “proposal distribution”, G :

$$\begin{aligned}\mathbb{E}_F \varphi(X) &= \mathbb{E}_G \left[\varphi(X) \cdot \frac{f(X)}{g(X)} \right] \\ &= \mathbb{E}_G [\varphi(X) \cdot w(X)]\end{aligned}$$

- G can be nearly anything*
 - *Some choices will be better than others

Importance Sampling

Option 1 (“Ideal” Monte Carlo):

- Generate $X_1, \dots, X_N \stackrel{iid}{\sim} \mathbb{F}$
- Estimate $\mathcal{I}$ with $\tilde{E} := \frac{1}{N} \sum \varphi(X_i)$
 - We often can't actually do this

Option 2 (Importance Sampling):

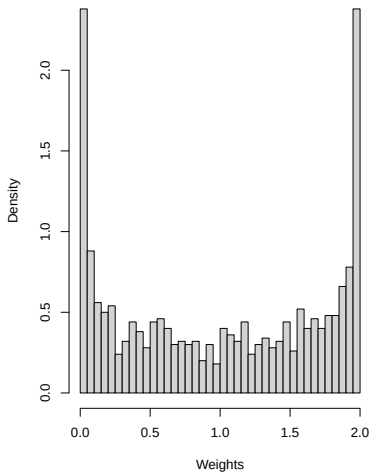
- Generate $X_1, \dots, X_N \stackrel{iid}{\sim} \mathbb{G}$
- Estimate $\mathcal{I}$ with $\hat{E} := \frac{1}{N} \sum w(X_i) \varphi(X_i)$

Example: Standard Normal

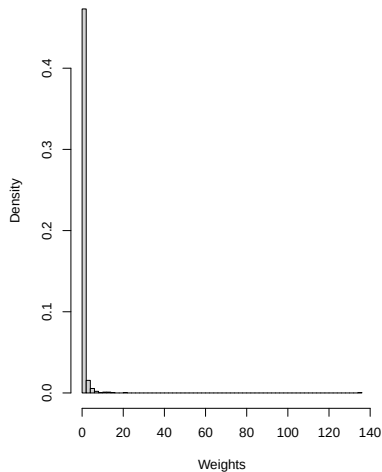
- f standard normal
- $\varphi(X) = X^2$
- Try some proposals:
 - $G_1 \sim N(0, 2^2)$
 - $G_2 \sim N(0, 0.5^2)$
- Use $M = 1000$ samples from proposal
 - $\hat{\mathbb{E}}_1 = 0.99, \hat{SD} = 1.97$
 - $\hat{\mathbb{E}}_2 = 1.10, \hat{SD} = 2.32$

Example: Standard Normal

$$G_1 = N(0, 2^2)$$



$$G_2 = N(0, 0.5^5)$$



Importance Sampling

- Nothing obviously wrong with the G_1 weights
- G_2 weights dominated by one large value
- We can make this difference precise

Effective Sample Size

- Effective sample size measures the sample efficiency of importance sampling relative to “ideal” Monte Carlo

$$ESS_* := N \cdot \frac{\mathbb{V} \tilde{E}}{\mathbb{V} \hat{E}}$$

Effective Sample Size

- If we choose a bad proposal, $\mathbb{V}\hat{E}$ will be large
→ ESS_* will be small
- If we choose a good proposal, $ESS_* \approx N$
- It's possible to choose $\mathbb{G}$ such that $\mathbb{V}\hat{E} < \mathbb{V}\tilde{E}$
 - $ESS_* > N$
 - This is why importance sampling is called a variance reduction technique

Effective Sample Size

- Problem: If we can't even evaluate $\hat{\mathbb{E}}$, we probably can't get $\mathbb{V}\hat{\mathbb{E}}$
- We need ways to estimate ESS_*

Effective Sample Size

- The most popular estimator for ESS_* is

$$ESS_2 := \frac{[\sum w(X_i)]^2}{\sum w(X_i)^2}$$

- This can be derived directly from ESS_* with $\varphi(X) = X$ and some very strong simplifying assumptions

Effective Sample Size

- This doesn't really look like it has anything to do with sample sizes, however:
 - If all weights are equal, $ESS_2 = N$
 - If there are k non-zero weights that are all equal, $ESS_2 = k$
- Problem: ESS_2 doesn't depend on φ
 - This is actually true for most ESS formulas

Aside: Self Normalization

- Sometimes, we only know the target density up to a proportionality constant
 - E.g. Simulating from posterior distributions
- Use this one simple trick:
- Divide each weight by the sum of the weights
 - Each weight is now in $(0, 1)$, and they sum to 1
 - Unknown proportionality constant cancels
- Self-normalized importance sampling

Aside: Self Normalization

- Write $\bar{w}_i$ for the i th normalized weight
 - Many ESS formulas assume that weights have been normalized
- Normalizing often simplifies formulas
- E.g. $ESS_2 = (\sum \bar{w}_i^2)^{-1}$

Aside: Self Normalization

- Unfortunately, normalizing also makes $\hat{\mathbb{E}}$ biased
 - Still consistent for $\mathcal{I}$
- For the rest of today, we will work with normalized weights

Effective Sample Size

- I said earlier that ESS_2 is based on strong assumptions about the weights
- Another way to motivate ESS_2 is based on the discrete uniform distribution
- Best case for IS is when all weights are equal
 - I.e. $\text{Weights} \sim \text{discrete uniform}$
- The farther we are from this ideal, the worse our ESS should be
- Different distances give us different ESS

Effective Sample Size

- ESS_2 comes from the $\mathcal{L}^2$ distance
- $\mathcal{L}^\infty$ gives us something quite different:

$$ESS_\infty := \frac{1}{\max \bar{w}_i}$$

Effective Sample Size

- We could also use $\mathcal{L}^1$

$$ESS_1 := -N \sum \bar{w}_i \cdot I(\bar{w}_i > 1/N) + N^+ + N,$$

where N^+ is the number of weights larger than $1/N$

- One last version of ESS, based on entropy instead of distances:

$$ESS_P := \exp \left[- \sum \bar{w}_i \log \bar{w}_i \right]$$

(Called perplexity)

Effective Sample Size

- All these formulas look quite different
- However, all satisfy the properties from earlier:
 - If all weights are equal, $ESS = N$
 - If there's k equal non-zero weights, $ESS = k$

Effective Sample Size

- Which ESS should we use?
- Some comparisons in the literature:
 - Martino et al. (2016) found that ESS_2 and ESS_∞ tend to bracket ESS_* in a small simulation study
 - Martino and Elvira (2025) found that it may be better to use $\mathcal{L}^p$ -based ESS with p as high as 8, depending on the scenario; also from small simulation studies

Adaptive Importance Sampling

- Instead of using ESS to measure the performance of a single proposal, use it to find a good proposal
- Adaptive importance sampling involves finding an optimal proposal for a given problem
 - Different ESS formulas give us different measures of optimality
- Perform optimization using stochastic approximation

Stochastic Approximation

- Stochastic approximation was proposed by Robbins and Monro in 1951 for finding roots of functions with noise
- The next year, Kiefer and Wolfowitz pointed out that this could find roots of gradients
 - Optimization of functions with noise

Stochastic Approximation

- A root of $f(x)$ is a fixed point of $x + f(x)$
 - Use fixed point iteration for root finding

$$x_{n+1} = x_n + f(x_n)$$

- Robbins and Monro showed that this recursion still works when you replace $f(x_n)$ with $\hat{f}(x_n) = f(x_n) + \varepsilon$
- Also need a dynamic step size

Stochastic Approximation

- The main reason we're interested in this method is to take f as a gradient
 - Now, we're doing optimization
- Many functions of interest don't have analytical gradients
- Kiefer and Wolfowitz (1952) suggested using a finite difference approximation

Stochastic Approximation

- Fixed-point iteration for root finding

$$x_{n+1} = x_n + f(x_n)$$

Stochastic Approximation

- Robbins and Monro (1951)

$$x_{n+1} = x_n + a_n \hat{f}(x_n)$$

Stochastic Approximation

- Robbins and Monro (1951) for optimization

$$x_{n+1} = x_n + a_n \widehat{\nabla} f(x_n)$$

Stochastic Approximation

- Kiefer and Wolfowitz (1952)

$$x_{n+1} = x_n + a_n \left[\frac{\hat{f}(x_n + c_n) - \hat{f}(x_n - c_n)}{2c_n} \right]$$

Stochastic Approximation

- Disclaimer: The theory and practice of stochastic approximation has developed since the 50s
- E.g. Stochastic gradient descent, Polyak-Ruppert averaging, projection, ...

Adaptive Importance Sampling

- Let's combine the idea of effective sample size with stochastic approximation
1. Choose an ESS measure and a parametric family of proposals
 2. Use stochastic approximation to maximize the chosen ESS
 3. Profit

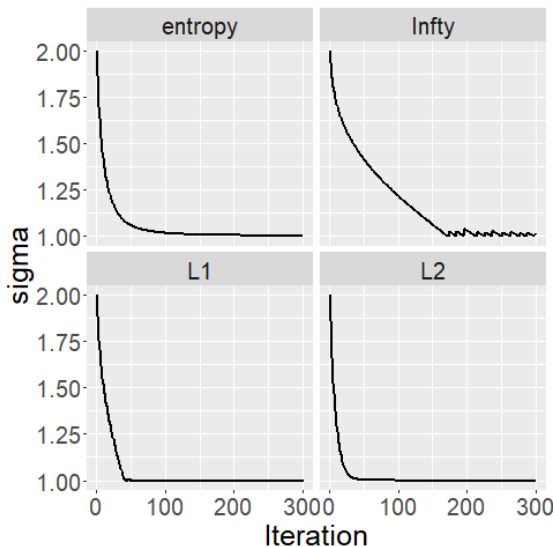
Example: Normal Scale Family

- Let's revisit the simple example from earlier
- f standard normal
- Proposal family: $\{N(0, \sigma^2)\}$

Example: Normal Scale Family

- Rather than choosing one ESS, I want to compare them
- Use stochastic approximation to optimize over $\sigma > 0$.

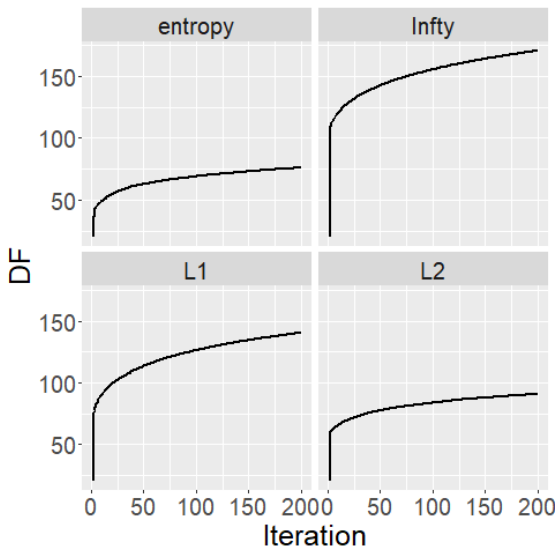
Example: Normal Scale Family



Example: T Degrees of Freedom

- Next, let's do something a little more interesting
- f standard normal
- Proposal family: $\{T_n : n > 0\}$

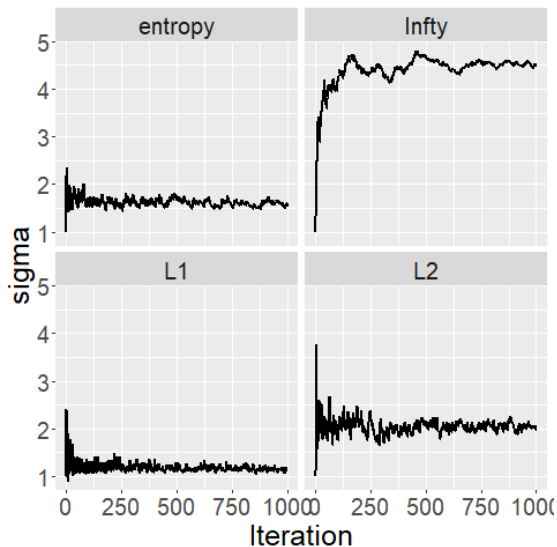
Example: T Degrees of Freedom



Example: T from Normal

- Finally, let's end on a weird example
- $f \sim \text{Cauchy}$
- Proposal family: $\{N(0, \sigma^2)\}$

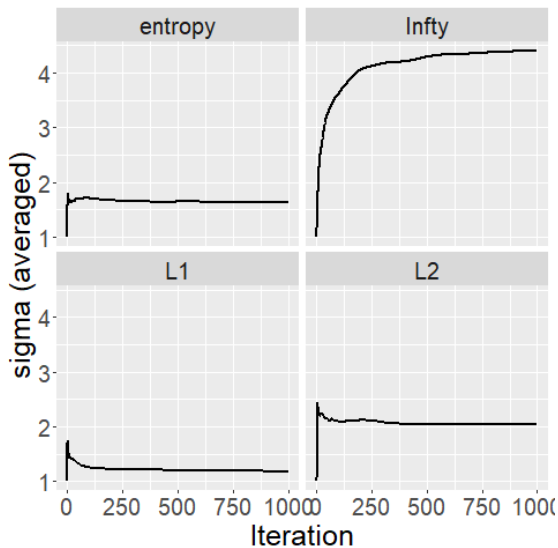
Example: T from Normal



Example: T from Normal

- We might be able to reduce this variability by carefully tuning SA
- Easier to use Polyak-Ruppert Averaging
 - Replace process with cumulative average

Example: T from Normal



Discussion

- Key takeaways:
 - You don't have to guess a great proposal distribution
 - There are different ways to measure effective sample size
- Ongoing work:
 - Guidance on which ESS to use when
 - Applications to latent variable problems

Collaborators

- Payman Nickchi (UBC)
- Richard Lockhart (SFU)

Thank You

Some References

- Kiefer, J. and Wolfowitz, J. (1952). Stochastic estimation of the maximum of a regression function. *The Annals of Mathematical Statistics*, 23(3).
- Martino, L. and Elvira, V. (2025). Effective sample size approximations as entropy measures. *Computational Statistics*, 40.
- Martino, L., Elvira, V., and Louzada, F. (2016). Alternative effective sample size measures for importance sampling. *2016 IEEE Signal Processing Workshop (SSP)*.
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